GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

NPTEL & IIT Professor Recommended GATE ECE Roadmap

Structured Subject Hierarchies, Lecture Mapping & Preparation Strategy

Curriculum: NPTEL / SWAYAM IIT Courses • **Scope:** Full GATE ECE Syllabus • **Timeline:** 12-Month Plan

1. Recommended NPTEL Courses & IIT Faculty References

- **Foundational Tier (Months 1 - 3):**

• *Network Analysis:* Prof. Tapas Kumar Bhattacharya (IIT Kharagpur) - Focus on Laplace transform of circuits, state variables, and two-port conversions.
• *Signals & Systems:* Prof. K.S. Venkatesh (IIT Kanpur) - Focus on CTFT duality, convolution algebra, and discrete-time sampling.
• *Engineering Mathematics:* Prof. Jitendra Kumar (IIT Kharagpur) - Linear algebra, vector calculus, and complex residues.

- **Core Electronics Tier (Months 4 - 6):**

• *Semiconductor Devices:* Prof. Nandita Dasgupta (IIT Madras) - Energy band diagrams, quasi-Fermi levels, and subthreshold slope.
• *Analog Electronic Circuits:* Prof. Nagendra Krishnapura (IIT Madras) - Negative feedback analysis, op-amp stability, and Miller compensation.
• *Digital Circuits:* Prof. S. Srinivasan (IIT Madras) - State machines, hazard detection, and setup/hold time analysis.

- **Advanced Systems Tier (Months 7 - 9):**

• *Control Systems:* Prof. M. Gopal (IIT Delhi) - Root locus geometry, Nyquist encirclement rules, and state feedback pole placement.
• *Communication Systems:* Prof. Surendra Prasad (IIT Delhi) - Digital constellations, matched filter derivation, and Shannon limits.
• *Electromagnetic Waves:* Prof. R.K. Shevgaonkar (IIT Bombay) - Smith chart impedance matching, waveguide modes, and antenna arrays.

2. 12-Month Preparation Schedule & Revision Cycle

- **Phase I (Comprehensive Theory & Formula Notebook):** Complete all 10 subjects sequentially alongside standard textbook exercise problems.
- **Phase II (Subject-Wise PYQ Solving):** Solve 30 years of GATE ECE previous year questions twice. Categorize mistakes into conceptual gaps vs calculation slips.
- **Phase III (Full-Length Mock Tests & Error Logging):** Attempt 15 - 20 full-length 3-hour mock exams in the official 9:30 AM or 2:30 PM slot. Maintain an active error logbook.